

Report

Jean Tirole (2012), *Overcoming Adverse Selection: How Public Intervention Can Restore Market Functioning*, *American Economic Review* 102(1): 29–59

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1 Introduction

Tirole asks a question that the 2007–09 crisis made important: when a market freezes because of adverse selection, can a government jump-start it, and if so, how? The answer is built on a single “twist” relative to standard mechanism design. The textbook designer (a monopolist screening a buyer, a regulator facing a firm) takes each agent’s reservation utility as a primitive: the agent always has a fixed outside option, and the participation constraint is $U(\theta) \geq U_0(\theta)$ with U_0 exogenous. Tirole’s point is that when the designer is a government acting before a private market reopens, this is no longer true. The outside option of a firm that declines the government’s offer is what it can get in the market, and, importantly, the market outcome depends on who declined—that is, on the mechanism itself. Reservation utilities are therefore endogenous to the mechanism. This two-way feedback between mechanism design and participation constraints is the paper’s conceptual core.

The economics that follow are interesting and a bit counterintuitive. Because the government moves first and skims off the worst (most “toxic”) assets, it cleans up the residual pool and so revives the very private market it is competing against—it “creates its own competition”. Anticipating the rebound, firms with good assets hold out for the market, so the government can never substitute fully for it: a non-comprehensive intervention with market rebound is optimal. Reviving a frozen market is necessarily expensive (the government loses money on every type it finances), yet—and this is the nice result—the mere existence of the later market imposes no welfare cost. The framework delivers a clean public-finance logic for when a freeze should, and should not, be undertaken.

In this report I will try to first summarize the key fundamentals of the model, the results and the possible extensions (section 2). In section 3 I will try to connect the paper’s contribution along what we have seen in the course. Finally, in section 4 I will end with some criticisms or comments of the paper.

2 The model

2.1 Primitives

All parties are risk neutral. A firm (seller) is cashless, protected by limited liability, owns a legacy asset, and has a new project to finance.

Legacy asset. The asset pays R_0 on success and 0 on failure. The success probability θ is the seller's private information, drawn from a continuous c.d.f. F on $[0, 1]$ with density f . Tirole assumes F is log-concave (decreasing reverse hazard rate $f(\theta)/F(\theta)$), the standard regularity condition that keeps the screening problem well behaved.

New project. The project costs I . If the entrepreneur behaves it yields verifiable income R_1 and private benefit $b > 0$; if she misbehaves it yields no income but a large private benefit $B > b$ and has negative social value ($B < I$). Define the social surplus of the project

$$S = R_1 + b - I.$$

Behaving requires leaving the entrepreneur a stake $B - b$, so the pledgeable income is only $R_1 - (B - b)$. Three assumptions organize the analysis:

Assumption 1 (Positive NPV). $S > 0$.

Assumption 2 (Scope for credit rationing). Without the legacy asset the project cannot be financed: $R_1 - (B - b) < I$, i.e. $S - B < 0$.

Assumption 3 (Collateral may enable financing). For the best types collateral restores financing under symmetric information: $R_0 + (S - B) > 0$.

The moral-hazard wedge B is what makes the firm credit-constrained: it cannot raise I on the project alone and must sell (a stake in) the legacy asset, whose quality only it knows.

Government and welfare. The government faces a shadow cost λ of public funds and maximizes expected gross social surplus

$$W = \mathbb{E}[U(\theta)] + \pi - (1 + \lambda)D,$$

where D is the deficit, $U(\theta)$ the seller's utility and π the buyers' profit. Since competitive buyers break even ($\pi = 0$) in equilibrium, letting $x(\theta) \in \{0, 1\}$ indicate financing,

$$W = (1 + \lambda) \mathbb{E}[\theta R_0 + x(\theta)S] - \lambda \mathbb{E}[U(\theta)].$$

The first term is efficiency (assets plus financed surplus); the second is the social cost of the rents the government must concede, weighted by λ . Basically, a government with $\lambda > 0$ is reluctant to leave large rents to firms—this is the tension that drives the whole design problem.

Timing and reservation utility. (1) The seller learns θ . (2) The government commits to a mechanism $C_g(\cdot)$. (3) Each seller accepts or declines. (4) Competitive buyers offer menus to the remaining sellers. (5) Sellers choose. (6) Payoffs realize. Laissez-faire simply deletes stages (2)–(3). For the leading “fleeting opportunity” case the autarky payoff of a seller who trades with nobody is

$$U_0(\theta) = \theta R_0.$$

2.2 The stripped-down version (buybacks only)

To isolate the mechanism, set $R_0 = 1$, suppose the entire project return belongs to the firm (no pledgeability), and take $\theta \sim U[\gamma, 1]$ with $\gamma > 0^1$. A seller of type θ who sells the asset at price p

¹The author will relax these assumptions and show that results mostly pass through and are general. I will discuss this in next subsection.

and finances the project obtains $p + S$; keeping the asset (autarky) is worth θ . Hence low types sell and high types keep:

$$\text{seller sells} \iff \theta < p + S =: \theta^*.$$

Competitive buyers break even at the truncated mean of what they buy:

$$p = \mathbb{E}[\theta \mid \theta < p + S] = \frac{1}{2}(\gamma + p + S) \implies p = \gamma + S, \quad \theta^* = \gamma + 2S.$$

This is Akerlof in two lines, and it delivers the discontinuity at the heart of a freeze:

- If $\gamma + S < I$ the break-even price is below I and **there is no trade**;
- If $\gamma + S \geq I$, all types $\theta < \gamma + 2S$ trade at $p = \gamma + S$.

With $S < I$ and $\gamma = 0$ the market fully collapses; a small piece of good news (a rise in γ) can “restart” it. Conversely, a small piece of bad news pushes γ below $I - S$ and freezes a previously active market—thus, there is discontinuous response to a marginal shock.

Adding the government (positive side). Suppose the government posts a price $p > I$. Were there no private market, all types $\theta < p + S$ would sell to it. But once those types are gone, the residual pool is better, so a private market reopens at a higher price—and anticipating this, the low types would rather hold out. The equilibrium reconciles the two: government and market prices are equated at p , types $\theta < p - S$ sell to the government, and an interval of intermediate types trades in the rebounding private market. The government loses money²; private buyers break even. This is the “creates its own competition” mechanism in its simplest form. Since the posted price pins down the financing cutoff one-for-one ($\theta^* = (p + S)/R_0$), choosing the price is the same as choosing θ^* , which is how the next step frames the government’s problem.

When does the government intervene? (normative side) With cutoff θ^* and shadow cost λ , intervening to finance the marginal type is worthwhile only if the efficiency gain beats the extra inframarginal rents:

$$\lambda F(\theta^*) < (1 + \lambda) f(\theta^*) S \iff \lambda < 1 \quad (\text{uniform case}).$$

So on an active market a government with $\lambda > 1$ stays out. But if the market freezes (γ falls below $I - S$), it is optimal to let it rebound to its minimum volume provided

$$(1 + \lambda) S > \lambda \frac{I + S - \gamma}{2},$$

i.e. provided $I - \gamma$ is not too far above S . The same government switches from laissez-faire to intervention precisely when bad news tips the market over the “bad cliff”.

2.3 The general model: what the richer contract space changes

The buyback case has a single instrument: a price. The government either buys the whole asset or it does not, and the only sorting that happens is through who chooses to sell. The general model adds a genuinely new instrument. Because part of the legacy return is now contractible, the government (or the market) can finance the firm while letting it retain a stake y in the legacy asset’s success—“skin in the game.” A financed type θ then either shirks and gets B , or works and gets $b + \theta y$, so its rent is

$$V(\theta y) = \max\{B, b + \theta y\}.$$

²The government and market prices are equated in equilibrium, so the pooled low types it buys ($\theta < p - S$) are paid a price above their fundamental value θR_0 —hence it loses money on them.

The key point is that this stake is itself a screening device: since $b + \theta y$ increases in θ , a good type values keeping a stake more than a bad type does. The government can therefore separate types by how much skin it asks them to keep, rather than only by price.

Formally one now works with incentive-compatible contracts and the rents they generate. Let $U_g(\theta)$ be type θ 's rent in the government mechanism and $U_m(\theta; \Theta_m)$ her rent in the market when the holdout set is Θ_m . An equilibrium partitions types into those who join the government, Θ_g , and those who go to the market, $\Theta_m = \Theta \setminus \Theta_g$, with $U_g(\theta) \geq U_m(\theta; \Theta_m) \Rightarrow \theta \in \Theta_g$ and the reverse for Θ_m . The government picks an IC mechanism (a rent profile U_g) to maximize W subject to this being the equilibrium of the fixed-point problem—exactly the endogenous participation constraint of the introduction.

The screening stake reshapes the form of the optimum (figures 3-4 and Proposition 7). The worst types are still bought out outright—a flat rent $U(\theta) = B$, no skin in the game—but better types are now financed while keeping a graduated stake, so their rent rises along $b + \theta y$. This is the **pecking order** (buy the worst, recapitalize the middle, leave the best), and now it is not assumed but derived: it is what optimal screening through revenue-sharing looks like. For the same reason the optimal intervention is more extensive than under buybacks: separating types by a retained stake is cheaper than separating them by outright purchase, so the government optimally reaches more types.

The remaining lessons of the buyback case survive the richer contract space essentially intact:

- (i) **Non-comprehensive intervention / rebound.** Short of the prohibitively costly policy of buying everything, the market always rebounds; the government reduces adverse selection enough to let it restart, but not so much as to make the intervention needlessly expensive.
- (ii) **Costly intervention.** The market already prices firms willingness to fire-sell, so rejuvenation is necessarily expensive: the government still loses money on every financed type.
- (iii) **No desire to shut the market down.** If the law forbids expropriation (firms get at least $U_0(\theta) = \theta R_0$), the voluntary-participation constraint can be made costless: the government would gain nothing from the power to ban the market, though the market deeply shapes how it intervenes.
- (iv) **When is intervention desirable?** A market failure does not by itself justify a bailout; the budgetary cost λ makes the government reluctant. Small bad news, however, can flip the optimum from laissez-faire to intervention as gains are there much higher.
- (v) **Intervention creates moral hazard.** The prospect of a bailout decreases ex-ante incentives to originate high-quality assets.

A by-product (Section III) connects to Rothschild–Stiglitz: the constrained-efficient allocation—highest surplus subject to seller IC and buyer break-even—is itself an equilibrium of the laissez-faire market, and the unique one surviving a “robust choice” refinement. The market left alone already implements the best incentive-feasible allocation; the government’s role is only to relax the adverse-selection constraint, at a fiscal cost.

2.4 Extensions

Tirole proposes then several extensions; I summarize them:

- *Ex-ante asset quality.* Endogenizing θ via costly effort confirms insight (vi): the anticipated rescue lowers the marginal return to originating good assets, so the intervention naturally creates moral hazard. With commitment the optimal scheme would also internalize this effort margin.
- *Fungibility (Myers–Majluf).* If legacy and project payoffs cannot be told apart, contracts reward one vs. two successes; the flat bottom of the constrained optimum (figure 4 in the paper) tilts upward and is no longer constant but the logic is mostly unchanged.
- *Two-sided adverse selection.* Private information on the new project too (i.e. $S(\theta)$) makes screening two-dimensional and hard, but under perfect correlation it collapses to one dimension and the analysis goes through³.
- *A small cost of intervention* pins down a unique implementation: the government rescues the bottom $[\theta, \theta_g]$ and leaves $[\theta_g, \theta^*]$ to the market, maximizing the size of the free market by clearing out the worst types.

3 Placement in the course

The paper sits mostly at the intersection of the four blocks we covered in class. Reading it was a good way to see how they fit together. I will try to connect them by parts:

Revelation principle and the uninformed-principal logic (Myerson 1982). The government’s problem is a mechanism-design problem, and the revelation principle lets us restrict attention to direct, incentive-compatible mechanisms $(U_g(\cdot), C_g(\cdot))$ —this is what licenses working with rent profiles rather than arbitrary contracts. What the paper changes is the participation side. In the canonical formulation the individual-rationality constraint is $U(\theta) \geq U_0(\theta)$ with U_0 fixed; here it becomes $U_g(\theta) \geq U_m(\theta; \Theta_m)$ with the right-hand side a function of the mechanism. The clean separation of IC and IR that we used throughout breaks down, and the designer’s problem becomes a fixed point. This is the single most important “twist” on the basics of the course that this paper brings.

Adverse selection and market breakdown. The stripped-down model is the Akerlof common-values market: a seller’s payoff depends on the type of whoever ends up holding the asset, the break-even price is a truncated mean, and incentive compatibility acts as a quantity/sorting constraint that generates a cross-type externality—precisely the externality that, under common values, can break decentralization (the Prescott–Townsend / Rothschild–Stiglitz theme from the Akerlof lecture). Tirole’s Section III result, that the constrained-efficient allocation is the unique refined market equilibrium, is the positive counterpart to that lecture’s existence and decentralization discussion.

Competing mechanisms / multiple principals. The government and the market are two principals contracting with the same population, so the paper belongs to the competing-mechanisms world of the course (Myerson 1982 competing hierarchies; Attar–Mariotti–Salanié on non-exclusive competition for lemons). But the placement is instructive precisely through a contrast: that literature has the principals move simultaneously, which is exactly why reservation utilities there are not mechanism-dependent and can be treated as exogenous. Tirole’s sequential

³A doubt however is what happens if $S(\theta)$ is exponentially increasing in its argument...

(Stackelberg) timing is what makes the outside option respond to the leader’s mechanism. So the paper can be read as the “competing-mechanisms” problem with the one ingredient the standard models assume away.

Moral hazard. Adverse selection is not the whole story: the credit constraint that forces the firm to sell its asset comes from a moral-hazard wedge $B - b$ on the new project (Assumption 2), and the limited pledgeability $R_1 - (B - b)$ is straight out of the Holmström–Tirole “skin in the game” logic of the moral-hazard part of the course. Insight (vi)—intervention distorting ex-ante origination—is itself a moral-hazard result put on top of the screening problem. The main results of the paper, however, focus on the adverse selection problem, which is far more important in the application.

4 Critical comments

In the end, I annotate some comments and doubts about the paper:

1. **Adverse selection as the cause of the freeze.** The whole apparatus traces freezes to asymmetric information about asset quality. Tirole is honest (Section V) that 2007–09 also reflected fire-sale / cash-in-the-market pricing, funding runs, regulatory forbearance, counterparty risk and ambiguity aversion. A model with no leverage, no rollover risk and no interconnectedness cannot speak to the run dynamics that arguably mattered most for Lehman/Bear. The paper’s value is conditional: if the binding friction is adverse selection, here is the optimal response. That said, the real contribution seems fundamentally methodological—endogenous, mechanism-dependent participation constraints—and this carries over to any sequential, multi-principal setting under adverse selection, well beyond the bailout story. That is what makes the paper matter beyond this critique.
2. **Commitment.** The analysis assumes the government commits at stage 2 to a mechanism it will not later revise. This is quite load-bearing: the clean timing that makes reservation utilities respond to the mechanism (rather than the reverse) requires the government to move first and stick to it. Tirole’s own moral-hazard extension shows the bite—with commitment the government would choose a smaller, cheaper intervention, but that policy is time inconsistent, since ex post it wants to raise the price back to the level implied by (2); anticipating the revision, firms behave as if there were no commitment, and the commitment gain evaporates. Real bailouts, renegotiated under political pressure (the drawn-out Japanese recapitalizations), look much more like the no-commitment case. Relaxing commitment fully, however, seems for future work.
3. **Specification of the outside option.** Several headline results rest on the fleeting-opportunity payoff $U_0 = \theta R_0$ together with no-expropriation. Tirole claims (fn. 12) the analysis extends to the less-urgent case $U_0 = \theta(R_0 + \delta S)$; I’d have liked to see whether the ‘no gain from shutting the market’ conclusion is equally clean there, since a higher outside option changes what the government must concede.
4. **On the positive side—and worth saying.** The pecking-order prediction (buy the worst, recapitalize the middle, leave the best) rationalizes the observed drift of TARP from outright toxic-asset purchases toward recapitalization (e.g. equity injections), and the “government must overpay” conclusion gives a coherent, if politically awkward, answer to the contemporaneous debate (Bebchuk, Krugman, Sachs vs. the “assets are undervalued, we

might even profit” view). The model articulates exactly why the optimistic view was wrong: a successful purge raises residual asset prices and makes good firms hold out, so the scheme must lose money. That is a counterintuitive but durable insight.

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